

About

This project showcases data visualization and performance analysis through an interactive Power BI dashboard designed to evaluate portfolio performance and attribution. The dataset includes benchmark index data and portfolio data derived from an investment fund's performance.

Key Components

- Data Processing: Data cleaning and transformation were conducted using Excel and Power Query.
- Calculations & Analysis: Implemented using DAX, including portfolio weight calculations, the Brinson attribution model for performance attribution, and key rate duration (KRD) analysis for fixed-income exposure.
- Dashboard Structure: The dashboard features five key pages:
 1. Summary – Provides an overview of portfolio performance.
 2. Equity Analysis – Performs attribution analysis for equity holdings.
 3. Fixed Income Analysis – Breaks down fixed-income asset performance by sector and risk factors.
 4. Risk Analysis – Assesses portfolio risks using metrics such as volatility, tracking error, and value at risk (VaR).
 5. Scenario Analysis – Examines portfolio performance under various economic conditions.

Limitations & Future Improvements

- Data Constraints: The dataset spans only nine months with monthly data points, limiting certain risk measurements such as volatility trends and drawdown analysis.
- Scope: The dataset includes only Canadian equity and fixed-income assets. Expanding the dataset to include global markets, additional asset classes, and a longer time horizon would enhance the depth and accuracy of the analysis.

Portfolio Summary 2024

Jan Feb Mar Apr May Jun Jul **Aug** Sep



Select a Date, Return Type, and Securities Type to view performance

MoM

YTD

EQUITY

FIXED INCOME SECURITIES



Portfolio Value (Previous Month) \$13,960,231		Portfolio Value (Current) \$14,107,874		Portfolio Value Change \$147,643	
Portfolio Return 0.70%		Benchmark Return 0.78%		Active Return -0.09%	
Portfolio Volatility 1.91%	Tracking Error 0.39%		Information Ratio -0.22		# of Trades 4

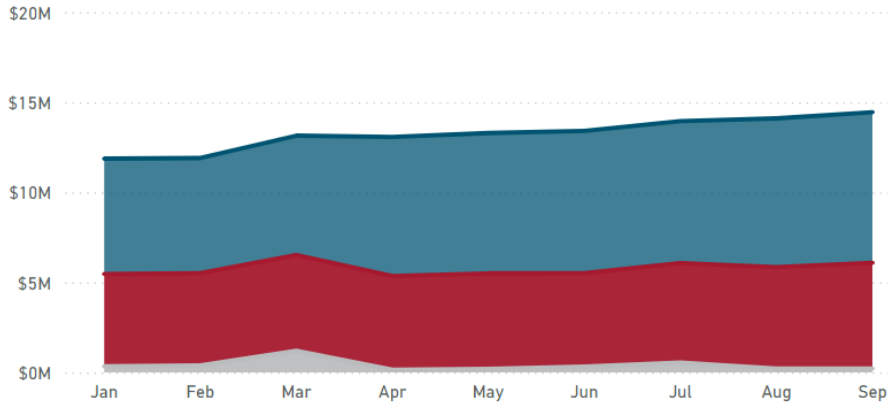
Note: Risk metrics are calculated using month-over-month (MoM) data, based on the most recent 6 months, and are not annualized.

Portfolio Composition

Value (\$)

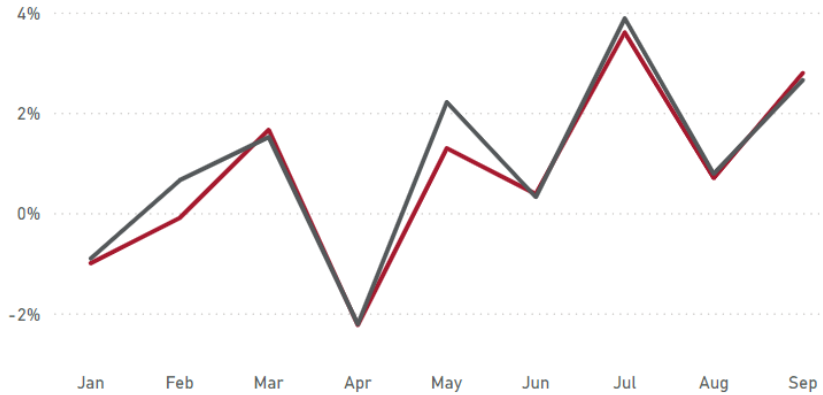
Weight (%)

● Cash & Cash Equivalents ● Equity ● Fixed Income



Portfolio Return vs. Benchmark Return

● Portfolio ● Benchmark



Equity

Fixed Income

2024

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

MoM Portfolio...

3.75%

MoM Benchmark...

3.43%

MoM Active Return

0.32%

YTD Portfolio...

4.14%

YTD Benchmark...

4.65%

YTD Active Return

-0.51%

Tracking Error

0.89%

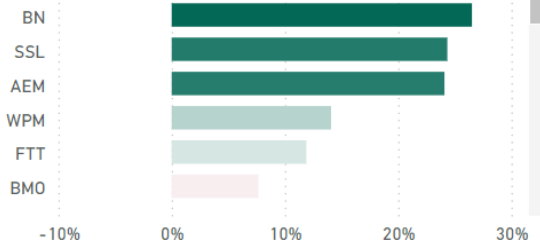
of Holdings

41

Sector/Security Return (%)

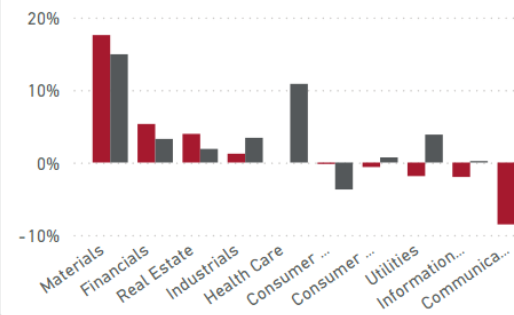
Dill up and down to view Return by Sector and Security

-8.60% 8.96% 26.52%

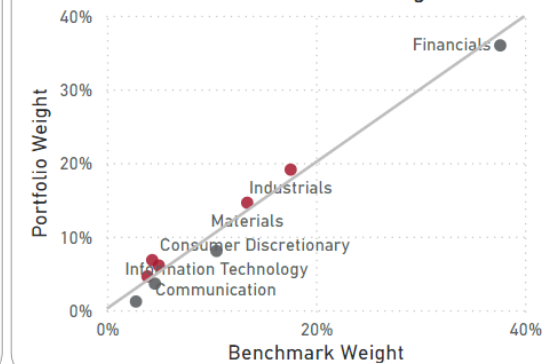


Portfolio Return vs. Benchmark Return by Sector

● Portfolio ● Benchmark

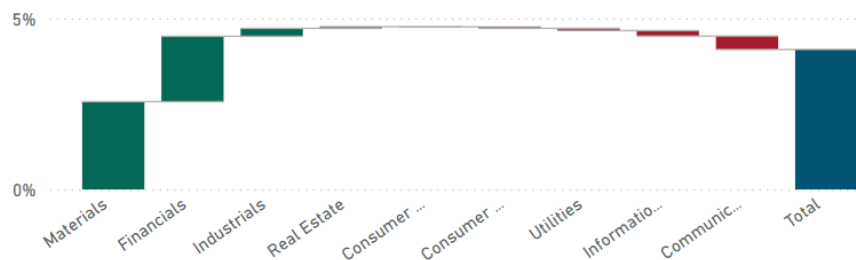


Portfolio vs. Benchmark Sector Weights



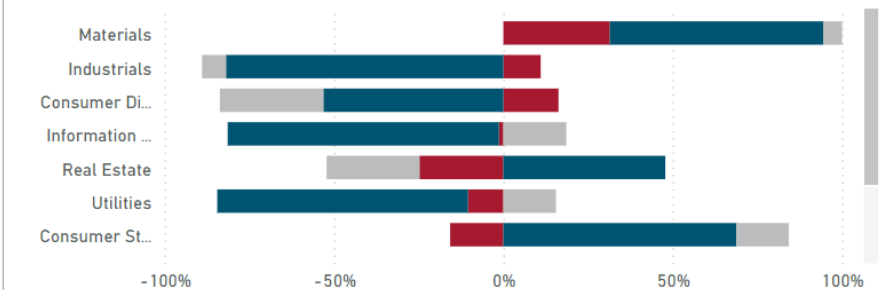
Return Contribution by Sector

● Increase ● Decrease ● Total



Active Return Attribution

● Allocation ● Selection ● Interaction



Equity

Fixed Income

All



Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

MoM Portfolio Return

-0.07%

MoM Benchmark Return

0.31%

MoM Active Return

-0.38%

Portfolio Duration

7.18

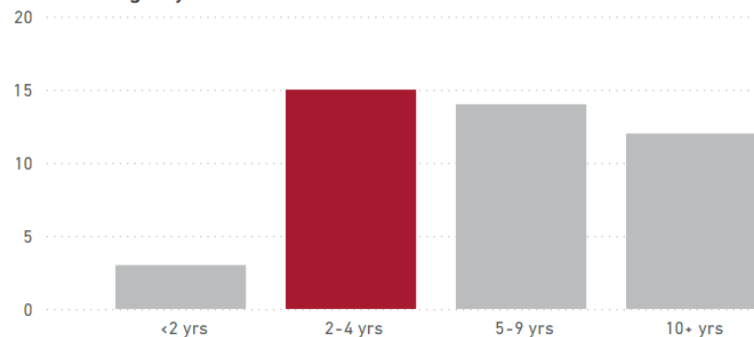
Benchmark Duration

7.18

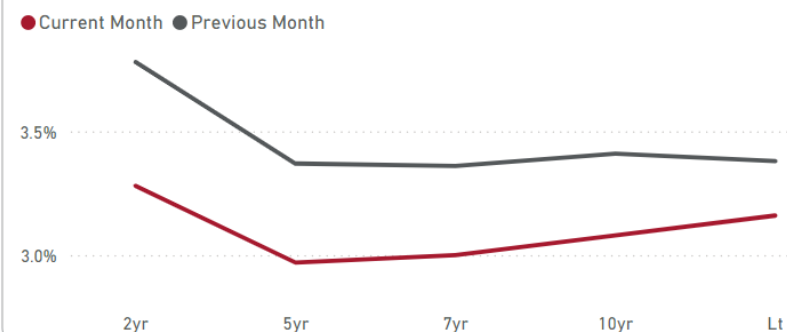
Duration Position

0.00

of Holdings by Duration



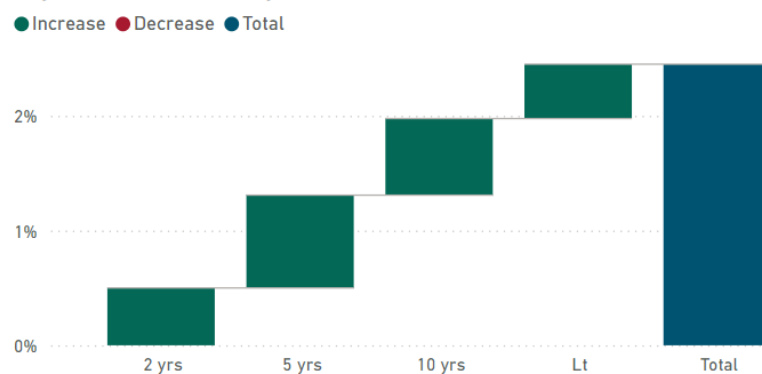
Term Structure of Interest Rate



Sector

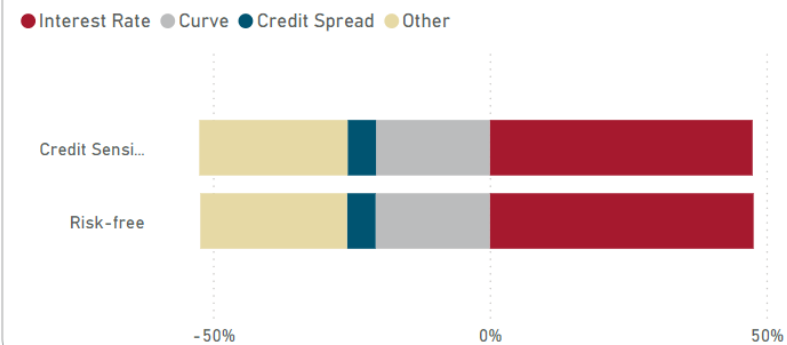
Factor

Key Rate Duration Analysis



Return Attribution by Factor

Dill up and down to view attribution by Sector Group and Sector



Risk Analysis

2024

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

EQUITY

FIXED INCOME SECURITIES

Portfolio Return

3.60%

Benchmark

3.88%

Portfolio Volatility

1.95%

Benchmark

1.78%

Portfolio Sharpe Ratio

-0.09

Benchmark

0.06

Active Return

-0.28%

Tracking Error

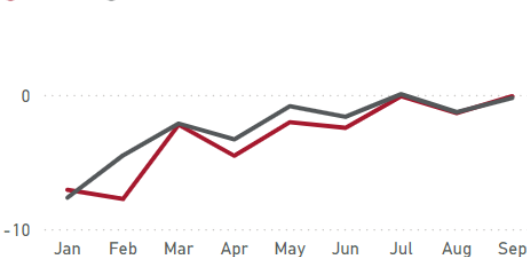
0.45%

Information Ratio

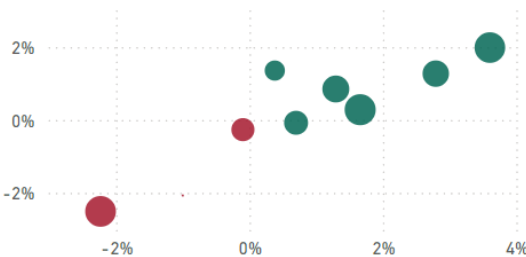
-0.63

Portfolio vs. Benchmark Sharpe Ratio

● Portfolio ● Benchmark

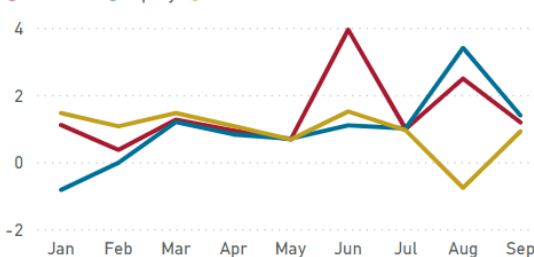


Correlation Between Equity & Fixed Income Portfolio

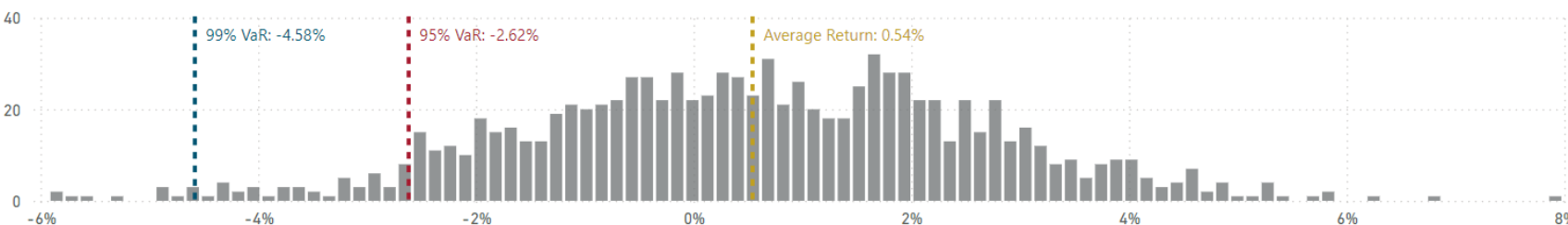


Beta Over Time

● Portfolio ● Equity ● Fixed Income



Portfolio VaR



Scenario Analysis

Results



Scenario	Equity Portfolio	Fixed Income Portfolio	Total Impact
Base Case	2.95%	0.10%	1.25%
Inflation	5.31%	-3.61%	0.11%
Recession	-4.36%	2.24%	-0.50%
Stagflation	0.66%	-0.88%	-0.23%
Total	4.55%	-2.14%	0.63%

Assumptions

Scenario	2yr Rate	5yr Rate	10yr Rate	30yr Rate	Growth Stock	Cyclical Stock	Defensive Stock
Base Case	-0.10%	-0.05%	0.00%	0.05%	5.00%	3.00%	1.00%
Inflation	0.70%	0.60%	0.50%	0.40%	12.00%	5.00%	3.00%
Recession	-0.50%	-0.40%	-0.30%	-0.20%	-10.00%	-5.00%	5.00%
Stagflation	0.30%	0.20%	0.10%	0.00%	-2.00%	0.00%	8.00%
Total	0.40%	0.35%	0.30%	0.25%	5.00%	3.00%	17.00%